

EXPERIMENT 8

Multiclass Classification for Student Performance Level Analysis

1. Problem Definition

The objective of this experiment is to classify students into multiple performance levels based on academic and behavioral attributes.

This is a **multiclass classification problem**, where:

- Class 0 - Low Performance
- Class 1 - Medium Performance
- Class 2 - High Performance

The goal is to correctly classify students and analyze model performance across multiple classes.

2. Data Understanding

The dataset contains **500 student records**.

Features include:

- Study Hours
- Attendance Percentage
- Assignment Score
- Internal Marks
- Participation
- Internet Access
- Previous Grade

Target variable:

- Performance_Level

Class Distribution

- Class 1 : **249**
- Class 0 : **247**
- Class 2 : **4**

3. Data Preprocessing

Steps Applied:

- Missing values removed
- Categorical variables encoded (Label Encoding)
- Feature scaling applied (StandardScaler)

4. Models Used

1. Decision Tree
2. Random Forest
3. Multinomial Logistic Regression
4. K-Nearest Neighbors (KNN)
5. XGBoost

5. Evaluation and Prediction

Since this is multiclass classification:

- Accuracy
- Precision (Macro Average)
- Recall (Macro Average)
- F1-Score (Macro Average)
- Confusion Matrix
- Accuracy alone is misleading

Model Performance Table

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
Decision Tree	0.87	0.9117	0.9116	0.9116
Random Forest	0.90	0.5998	0.6122	0.6060
Logistic Regression	0.97	0.6474	0.6599	0.6533
KNN	0.79	0.5272	0.5374	0.5303
XGBoost	0.95	0.6341	0.6463	0.6399

Confusion Matrix

Decision Tree

```
Confusion Matrix:
[[43  6  0]
 [ 7 42  0]
 [ 0  0  2]]
```

Random Forest

```
Confusion Matrix:
[[46  3  0]
 [ 5 44  0]
 [ 0  2  0]]
```

Multinomial Logistic Regression

```
Confusion Matrix:
[[48  1  0]
 [ 0 49  0]
 [ 0  2  0]]
```

K-Nearest Neighbors

```
Confusion Matrix:
[[39 10  0]
 [ 9 40  0]
 [ 0  2  0]]
```

XGBoost

```
Confusion Matrix:
[[47  2  0]
 [ 1 48  0]
 [ 0  2  0]]
```

- Class 2 has only **2 test samples**
- Many models fail to classify this class properly
- Undefined metric warnings appear → due to missing predictions

Interpretation:

- Models are biased toward Class 0 and 1
- Class 2 performance is unreliable

6. Model Comparison Analysis

1. Accuracy is MISLEADING

- Logistic Regression = **97% accuracy**
- But macro F1 = **0.65**

This means

- Model ignores minority class (Class 2)
- Still gets high accuracy

2. Decision Tree is Best

- Macro F1 = **0.91 (highest)**
- Balanced performance across classes

This is the ONLY model handling all classes properly

3. Random Forest Performs Worse than Expected

- Accuracy = 0.90 (looks good)
- F1 = 0.60 (bad)

Reason:

- Overfitting to majority classes

4. XGBoost is Strong but Not Best

- Accuracy = 0.95
- F1 = 0.64

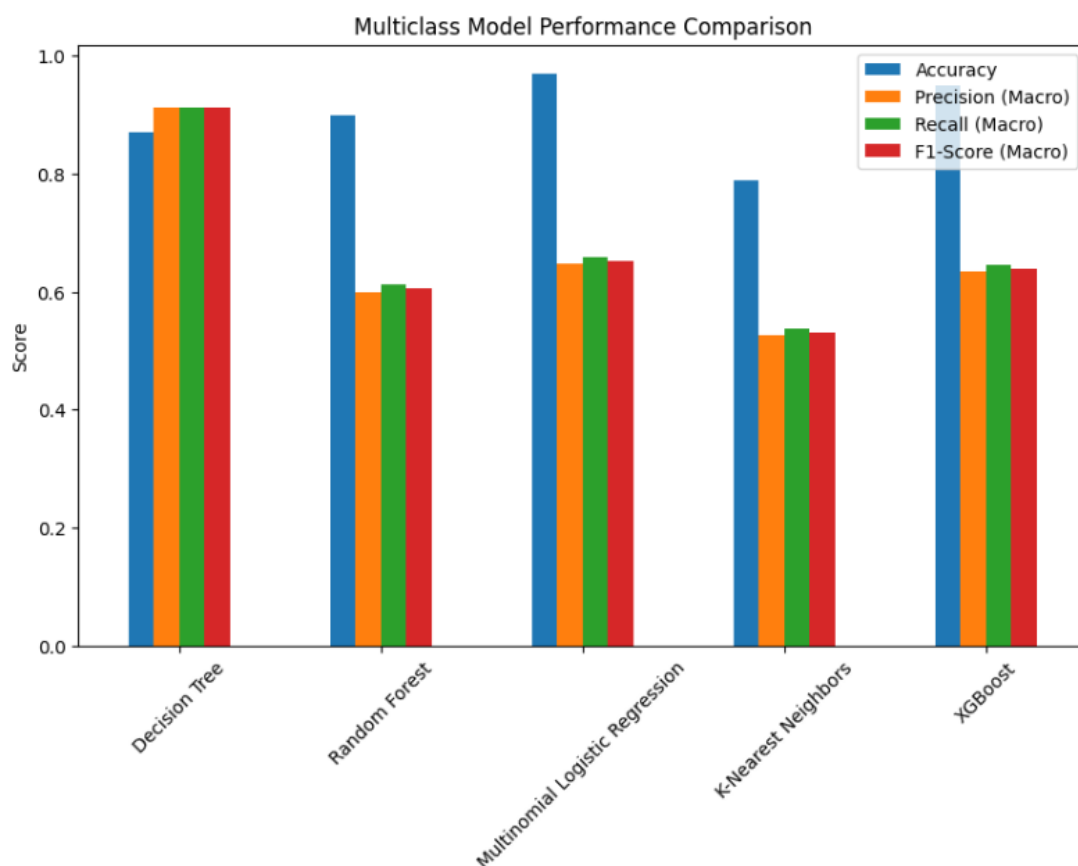
Still struggles with minority class

5. KNN is Weakest

- Lowest F1-score = 0.53
- Sensitive to scaling and data distribution

Visualization Analysis

- Decision Tree shows **balanced high bars across all metrics**
- Logistic Regression has high accuracy but lower macro scores
- KNN clearly underperforms



Best Model Selection

Decision Tree is the best model, because:

- Highest Macro F1-score (0.91)
- Balanced performance across all classes
- Handles minority class better